

Amazon Fresh Database

1. Introduction

This project focuses on designing and analyzing an **Amazon Fresh database** using SQL. The objective is to understand how relational databases can be created, managed, and queried to handle information related to customers, products, suppliers, orders, order details, and reviews.

The project covers important SQL concepts including **DDL, DML, primary keys, foreign keys, constraints, filtering, aggregation, joins, subqueries, window functions, and normalization**. Various SQL queries were developed to retrieve useful information and generate business-oriented insights from the database.

2. Analysis

2.1 Database and Table Management

The database was created as **Amazon**, with tables representing major business entities such as customers, products, suppliers, orders, order details, and reviews. Primary and foreign keys were considered to establish relationships between these tables.

The project also demonstrates the use of constraints. For example, the Customers table includes requirements for a primary key, non-null age, an age greater than 18, and uniqueness for customer names. These constraints help maintain data integrity.

2.2 Customer and Product Analysis

SQL queries were used to retrieve customers from a particular city and products belonging to a specific category such as **Fruits**. These queries demonstrate the use of the WHERE clause and filtering conditions to obtain relevant records.

2.3 Product and Inventory Management

New bakery products such as **Cream Bun, Butter Bun, and Fruit Bun** were inserted into the Products table. The project also includes an update to product stock quantity based on a specific Productid.

Supplier information was analysed by grouping products according to Supplier ID and calculating total stock. The supplier with the highest total stock was identified using GROUP BY, ORDER BY, and LIMIT.

2.4 Customer Spending Analysis

Customer spending was analysed using aggregate functions and joins. The total amount spent by each customer was calculated, and customers were ranked according to their spending using the RANK() window function.

The analysis also identifies customers whose total order amount exceeds **₹5,000**, which can help the business recognize high-value customers.

2.5 Order and Revenue Analysis

The project uses joins between the **Orders** and **Order Details** tables to calculate revenue at the order level. Grouping and aggregation are used to summarize order information.

Queries were also developed to identify customers placing orders during a particular period and to rank orders according to their total sales value.

2.6 Reviews and Ratings

A CHECK constraint was added to the Reviews table to ensure that ratings remain between **1 and 5**. The project also uses the HAVING clause to identify products whose average rating is greater than 4.

This provides a way to identify highly rated products and maintain the quality of review data.

2.7 Database Normalization

The Products table was considered for normalization into separate **Categories** and **Subcategories** tables. This approach reduces redundancy and improves the organization of product-related information.

The project also indicates the creation of relationships using foreign keys to maintain consistency between the normalized tables.

2.8 Subqueries and Business Insights

Subqueries were used to identify the top three products based on sales-related calculations. Additional queries were designed to identify customers without orders and analyze the concentration of Prime members by city.

The project also attempts to identify the three most frequently ordered product categories, providing useful information for understanding product demand.

3. Conclusion

The Amazon Fresh SQL project demonstrates how SQL can be used to **store, manage, retrieve, and analyse business data** effectively. The project covers fundamental and advanced SQL concepts, including table constraints, data manipulation, joins, aggregation, grouping, subqueries, ranking functions, and normalization.

The analysis can help an organization understand **customer spending, product performance, inventory levels, supplier stock, product ratings, and category demand**. Overall, the project provides practical experience in database management and shows how SQL queries can transform raw database records into useful business information.